

```

1 namespace CSharp;
2 class LoopTest
3 {
4     public static void WhileTest()
5     {
6         int i, n;
7         Console.WriteLine("Enter N:");
8         n = Convert.ToInt32(Console.ReadLine());
9         i = 1;
10        while (i <= n)
11        {
12            Console.WriteLine(i++);
13        }
14    }
15
16    public static void DoTest()
17    {
18        int i, n;
19        Console.WriteLine("Enter N:");
20        n = Convert.ToInt32(Console.ReadLine());
21        i = 1;
22        do
23        {
24            Console.WriteLine(i++);
25        } while (i <= n);
26    }
27
28    public static void ForTest()
29    {
30        int i, n;
31        Console.WriteLine("Enter N:");
32        n = Convert.ToInt32(Console.ReadLine());
33        i = 1;
34        for (i = 1; i <= n; i++)
35        {
36            Console.WriteLine(i);
37        }
38    }
39
40    public static void ForEachTest()
41    {
42        int i;
43        int[] a = { 1, 2, 3, 4 };
44        foreach (int e in a)
45        {
46            Console.WriteLine(e);
47        }
48    }
49
50    public static void BreakTest()
51    {
52        int i, n;
53        Console.WriteLine("Enter N:");
54        n = Convert.ToInt32(Console.ReadLine());
55        i = 1;
56        while (i <= n)
57        {
58            if (i == 4)
59                break;
60            Console.WriteLine(i++);
61        }
62    }
63
64    public static void ForEachTest()
65    {
66        int i, n;
67        Console.WriteLine("Enter N:");
68        n = Convert.ToInt32(Console.ReadLine());
69        i = 1;
70        while (i <= n; i++)
71        {
72            if (i == 2)
73            {
74                i++;
75                continue;
76            }
77            Console.WriteLine(i++);
78        }
79    }
80
81    public static void GotoTest()
82    {
83        int i, j, k, n;
84        Console.WriteLine("Enter N:");
85        n = Convert.ToInt32(Console.ReadLine());
86        for (i = 1; i <= n; i++)
87        {
88            for (j = 1; j <= n; j++)
89            {
90                for (k = 1; k <= n; k++)
91                {
92                    if (i == 2 && j == 3 && k == 4)
93                        goto outer;
94                    Console.WriteLine(" i={0} j={1} k={2}", i, j, k);
95                }
96            }
97        }
98        outer:
99        Console.WriteLine("End");
100    }
101
102    public static void ContinueTest2()
103    {
104        int i, n;
105        Console.WriteLine("Enter N:");
106        n = Convert.ToInt32(Console.ReadLine());
107        i = 0;
108        while (i < n)
109        {
110            Console.WriteLine(i);
111            if (i < 3)
112                continue;
113            i++;
114            Console.WriteLine(i);
115        }
116    }
117 }

```